Grade 7 · Unit 1 · Forces and Motion — Flashcards

PA STEELS 3.2.6-8.G, H, I, J, K

GR 7 · UNIT 1 | STEELS 3.2.6-8.H | #1

Force

A push or a pull on an object. Every force has a strength and a direction.

Gr 7 · Unit 1 · #1 · 3.2.6-8.H

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 | STEELS 3.2.6-8.H | #2

Mass

The amount of matter in an object. More mass means it takes more force to change the object's motion.

Gr 7 · Unit 1 · #2 · 3.2.6-8.H

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 | STEELS 3.2.6-8.H | #3

Frame of reference

The starting point and directions you choose to describe where things are and how they move. Example: "3 m east of the door."

Gr 7 · Unit 1 · #3 · 3.2.6-8.H

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 STEELS 3.2.6-8.H #4

Units

Agreed-on sizes for measuring, like meters for distance or newtons for force. Always write the unit with a number.

Gr 7 · Unit 1 · #4 · 3.2.6-8.H

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 STEELS 3.2.6-8.H #5

Why must you share your reference frame and units?

Your choices of reference frame and units are up to you. Others need to know them to understand your positions, directions, and data.

Gr 7 · Unit 1 · #5 · 3.2.6-8.H

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 STEELS 3.2.6-8.H #6

Net force (sum of forces)

All the forces on an object added together, counting their directions. Example: 5 N right and 3 N left make a net force of 2 N right.

Gr 7 · Unit 1 · #6 · 3.2.6-8.H

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 STEELS 3.2.6-8.H #7

Balanced forces

Forces on an object that cancel out, so the net force is zero. The object's motion does not change.

Gr 7 · Unit 1 · #7 · 3.2.6-8.H

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 STEELS 3.2.6-8.H #8

Unbalanced forces

Forces on an object that do not cancel out, so the net force is not zero. The object speeds up, slows down, or changes direction.

Gr 7 · Unit 1 · #8 · 3.2.6-8.H

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 STEELS 3.2.6-8.H #9

Newton's first law

If the forces on an object are balanced, its motion stays the same: an object at rest stays at rest, and a moving object keeps the same speed and direction.

Gr 7 · Unit 1 · #9 · 3.2.6-8.H

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 STEELS 3.2.6-8.H #10

Newton's second law

An object's change in motion depends on the net force and its mass. More force means a bigger change; more mass means a smaller change.

Gr 7 · Unit 1 · #10 · 3.2.6-8.H

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 STEELS 3.2.6-8.H #11

How does a bigger force affect an object's motion?

For the same object, a larger force causes a larger change in motion. Kicking a ball harder makes it speed up more.

Gr 7 · Unit 1 · #11 · 3.2.6-8.H

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 STEELS 3.2.6-8.H #12

How does mass affect the force needed to change motion?

The greater the mass, the greater the force needed for the same change in motion. A full cart is harder to start than an empty one.

Gr 7 · Unit 1 · #12 · 3.2.6-8.H

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 STEELS 3.2.6-8.G #13

Newton's third law

When two objects interact, they push or pull on each other with forces that are equal in strength and opposite in direction.

Gr 7 · Unit 1 · #13 · 3.2.6-8.G

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 STEELS 3.2.6-8.G #14

Collision

When two moving objects, or a moving and a still object, hit each other. Example: a car hitting a wall.

Gr 7 · Unit 1 · #14 · 3.2.6-8.G

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 STEELS 3.2.6-8.G #15

When a car hits a wall, which pushes harder?

Neither. By Newton's third law, the car pushes on the wall exactly as hard as the wall pushes on the car, in the opposite direction.

Gr 7 · Unit 1 · #15 · 3.2.6-8.G

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 STEELS 3.2.6-8.G #16

How does Newton's third law help design for collisions?

Each object in a collision gets an equal, opposite force. Engineers use this to design ways to reduce damage, like car bumpers or spacecraft shields against meteors.

Gr 7 · Unit 1 · #16 · 3.2.6-8.G

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 STEELS 3.2.6-8.I #17

Electric charge

A property of matter that causes electric forces. Objects with charge can push or pull on other charged objects.

Gr 7 · Unit 1 · #17 · 3.2.6-8.I

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 STEELS 3.2.6-8.I #18

Electric current

A flow of electric charge, like in a wire. Electric currents can cause magnetic forces, as in an electromagnet.

Gr 7 · Unit 1 · #18 · 3.2.6-8.I

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 STEELS 3.2.6-8.I #19

Electric force

A push or pull between objects with electric charge. Example: a charged strip of tape pulling on another strip.

Gr 7 · Unit 1 · #19 · 3.2.6-8.I

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 STEELS 3.2.6-8.I #20

Magnetic force

A push or pull caused by magnets or by electric currents. Example: two magnets snapping together.

Gr 7 · Unit 1 · #20 · 3.2.6-8.I

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 STEELS 3.2.6-8.I #21

Electromagnetic force

The name for electric and magnetic forces together. These forces can attract or repel.

Gr 7 · Unit 1 · #21 · 3.2.6-8.I

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 STEELS 3.2.6-8.I #22

Attract

To pull toward each other. Example: the opposite ends of two magnets attract.

Gr 7 · Unit 1 · #22 · 3.2.6-8.I

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 STEELS 3.2.6-8.I #23

Repel

To push away from each other. Example: the matching ends of two magnets repel.

Gr 7 · Unit 1 · #23 · 3.2.6-8.I

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 STEELS 3.2.6-8.I #24

What makes electric and magnetic forces stronger?

Bigger charges, bigger currents, or stronger magnets make the force stronger. So does moving the objects closer together.

Gr 7 · Unit 1 · #24 · 3.2.6-8.I

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 STEELS 3.2.6-8.I #25

How does distance affect electric and magnetic forces?

The farther apart the objects are, the weaker the force. Two magnets pull hard up close but barely at all when far apart.

Gr 7 · Unit 1 · #25 · 3.2.6-8.I

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 STEELS 3.2.6-8.I #26

Electromagnet

A magnet made by running electric current through a coil of wire. It can be turned on and off.

Gr 7 · Unit 1 · #26 · 3.2.6-8.I

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 STEELS 3.2.6-8.I #27

How can you make an electromagnet stronger?

Add more turns of wire to the coil or use more current. Both make its magnetic force stronger.

Gr 7 · Unit 1 · #27 · 3.2.6-8.I

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 | STEELS 3.2.6-8.I | #28

Electric motor

A device that uses electric and magnetic forces to make something spin. Stronger or more magnets can make it spin faster.

Gr 7 · Unit 1 · #28 · 3.2.6-8.I

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 | STEELS 3.2.6-8.I | #29

Generator

A device that uses magnets and motion to produce an electric current.

Gr 7 · Unit 1 · #29 · 3.2.6-8.I

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 | STEELS 3.2.6-8.J | #30

Gravitational force (gravity)

A force that pulls any two masses toward each other. Gravity is always attractive; it never pushes.

Gr 7 · Unit 1 · #30 · 3.2.6-8.J

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 | STEELS 3.2.6-8.J | #31

How does mass affect gravitational force?

The more mass the objects have, the stronger the gravity between them. That is why the sun's pull holds the planets.

Gr 7 · Unit 1 · #31 · 3.2.6-8.J

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 STEELS 3.2.6-8.J #32

Why don't you feel gravity between two pencils?

There is gravity between any two masses, but it is very small unless one object has a large mass, like Earth or the sun.

Gr 7 · Unit 1 · #32 · 3.2.6-8.J

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 STEELS 3.2.6-8.J #33

Orbital period

The time an object takes to go once around another object. Earth's orbital period around the sun is about 365 days.

Gr 7 · Unit 1 · #33 · 3.2.6-8.J

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 STEELS 3.2.6-8.K #34

Force at a distance

A force that acts without objects touching. Electric, magnetic, and gravitational forces act at a distance.

Gr 7 · Unit 1 · #34 · 3.2.6-8.K

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 STEELS 3.2.6-8.K #35

Field

A region in space around an object where it can push or pull on other objects without touching them.

Gr 7 · Unit 1 · #35 · 3.2.6-8.K

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 STEELS 3.2.6-8.K #36

Electric field

The field around a charged object. Example: a charged balloon pulls on a tiny pith ball before they touch.

Gr 7 · Unit 1 · #36 · 3.2.6-8.K

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 STEELS 3.2.6-8.K #37

Magnetic field

The field around a magnet or electric current. Example: a magnet moves a paper clip before they touch.

Gr 7 · Unit 1 · #37 · 3.2.6-8.K

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 STEELS 3.2.6-8.K #38

Test object

An object used to map a field by seeing how it is pushed or pulled at different spots. A small charged object can map an electric field.

Gr 7 · Unit 1 · #38 · 3.2.6-8.K

✂ cut on dashes · fold on dots

GR 7 · UNIT 1 STEELS 3.2.6-8.K #39

What evidence shows a field exists?

Objects move or change without touching. Example: two charged strips of tape push apart while still separated.

Gr 7 · Unit 1 · #39 · 3.2.6-8.K

✂ cut on dashes · fold on dots

Grade 7 · Unit 2 · Energy — Flashcards

PA STEELS 3.2.6-8.L, M, N, O, P

GR 7 · UNIT 2 | STEELS 3.2.6-8.L | #1

Energy

The ability to cause change, such as making something move or warm up.

Gr 7 · Unit 2 · #1 · 3.2.6-8.L

✂ cut on dashes · fold on dots

GR 7 · UNIT 2 | STEELS 3.2.6-8.L | #2

Kinetic energy

The energy of motion (also called motion energy). Anything that is moving has kinetic energy.

Gr 7 · Unit 2 · #2 · 3.2.6-8.L

✂ cut on dashes · fold on dots

GR 7 · UNIT 2 | STEELS 3.2.6-8.L | #3

How does mass affect kinetic energy?

At the same speed, more mass means more kinetic energy. Double the mass, double the kinetic energy. A tennis ball hits harder than a wiffle ball.

Gr 7 · Unit 2 · #3 · 3.2.6-8.L

✂ cut on dashes · fold on dots

GR 7 · UNIT 2 STEELS 3.2.6-8.L #4

How does speed affect kinetic energy?

Kinetic energy grows with the square of speed. Double the speed and the kinetic energy becomes 4 times as large.

Gr 7 · Unit 2 · #4 · 3.2.6-8.L

✂ cut on dashes · fold on dots

GR 7 · UNIT 2 STEELS 3.2.6-8.M #5

Particles

The tiny pieces, such as atoms, that all matter is made of. They are always moving.

Gr 7 · Unit 2 · #5 · 3.2.6-8.M

✂ cut on dashes · fold on dots

GR 7 · UNIT 2 STEELS 3.2.6-8.M #6

Temperature (in science)

A measure of the average kinetic energy of the particles in matter. Higher temperature means the particles move faster on average.

Gr 7 · Unit 2 · #6 · 3.2.6-8.M

✂ cut on dashes · fold on dots

GR 7 · UNIT 2 STEELS 3.2.6-8.M #7

Thermal energy

The total energy of the moving particles in a sample of matter. Adding thermal energy makes the particles move faster.

Gr 7 · Unit 2 · #7 · 3.2.6-8.M

✂ cut on dashes · fold on dots

GR 7 · UNIT 2 STEELS 3.2.6-8.M #8

Which way does thermal energy move on its own?

From hotter objects or places to colder ones. A hot drink warms your cold hands.

Gr 7 · Unit 2 · #8 · 3.2.6-8.M

✂ cut on dashes · fold on dots

GR 7 · UNIT 2 STEELS 3.2.6-8.M #9

Thermal energy transfer

The movement of thermal energy from one object or place to another, always from hotter to colder on its own.

Gr 7 · Unit 2 · #9 · 3.2.6-8.M

✂ cut on dashes · fold on dots

GR 7 · UNIT 2 STEELS 3.2.6-8.M #10

Insulation

Material that slows down thermal energy transfer. A Styrofoam cup or insulated box keeps things hot or cold longer.

Gr 7 · Unit 2 · #10 · 3.2.6-8.M

✂ cut on dashes · fold on dots

GR 7 · UNIT 2 STEELS 3.2.6-8.M #11

Solar cooker

A device designed to collect energy from sunlight and keep it inside to heat food. It maximizes thermal energy transfer to the food.

Gr 7 · Unit 2 · #11 · 3.2.6-8.M

✂ cut on dashes · fold on dots

GR 7 · UNIT 2 STEELS 3.2.6-8.M #12

Why define a design's criteria and constraints precisely?

The more clearly you define what the design must do and its limits, the more likely your solution will succeed.

Gr 7 · Unit 2 · #12 · 3.2.6-8.M

✂ cut on dashes · fold on dots

GR 7 · UNIT 2 STEELS 3.2.6-8.M #13

Why test a design solution?

Testing shows how well it works. You then change, or modify, the design based on the results to improve it.

Gr 7 · Unit 2 · #13 · 3.2.6-8.M

✂ cut on dashes · fold on dots

GR 7 · UNIT 2 STEELS 3.2.6-8.M #14

How do engineers evaluate solutions?

They use a step-by-step (systematic) process to check how well each solution meets the criteria and constraints.

Gr 7 · Unit 2 · #14 · 3.2.6-8.M

✂ cut on dashes · fold on dots

GR 7 · UNIT 2 STEELS 3.2.6-8.N #15

System

The group of objects or matter you choose to study together. Example: a cup of water and the ice in it.

Gr 7 · Unit 2 · #15 · 3.2.6-8.N

✂ cut on dashes · fold on dots

GR 7 · UNIT 2 STEELS 3.2.6-8.N #16

Are temperature and total energy the same thing?

No. A system's total energy at a given temperature depends on the types, states, and amounts of matter in it.

Gr 7 · Unit 2 · #16 · 3.2.6-8.N

✂ cut on dashes · fold on dots

GR 7 · UNIT 2 STEELS 3.2.6-8.N #17

What affects the energy needed to change a temperature?

The type of matter, the size (mass) of the sample, and the environment around it.

Gr 7 · Unit 2 · #17 · 3.2.6-8.N

✂ cut on dashes · fold on dots

GR 7 · UNIT 2 STEELS 3.2.6-8.N #18

Same energy added: does a small or large mass warm more?

The smaller mass of the same material warms more. A large mass needs more energy for the same temperature change.

Gr 7 · Unit 2 · #18 · 3.2.6-8.N

✂ cut on dashes · fold on dots

GR 7 · UNIT 2 STEELS 3.2.6-8.N #19

Do all materials change temperature the same way?

No. Samples of different materials with the same mass change temperature by different amounts when heated or cooled.

Gr 7 · Unit 2 · #19 · 3.2.6-8.N

✂ cut on dashes · fold on dots

GR 7 · UNIT 2 STEELS 3.2.6-8.N #20

Why does water get colder when ice melts in it?

Thermal energy moves from the warmer water into the colder ice. More ice takes more energy, so the water ends up colder.

Gr 7 · Unit 2 · #20 · 3.2.6-8.N

✂ cut on dashes · fold on dots

GR 7 · UNIT 2 STEELS 3.2.6-8.O #21

Energy transfer

When energy moves from one object or place to another. Example: a moving ball hits a still ball and makes it move.

Gr 7 · Unit 2 · #21 · 3.2.6-8.O

✂ cut on dashes · fold on dots

GR 7 · UNIT 2 STEELS 3.2.6-8.O #22

What happens when an object's kinetic energy changes?

Energy is transferred to or from the object. Some other change in energy always happens at the same time.

Gr 7 · Unit 2 · #22 · 3.2.6-8.O

✂ cut on dashes · fold on dots

GR 7 · UNIT 2 STEELS 3.2.6-8.O #23

Conservation of energy

Energy is not created or destroyed. It is transferred between objects or changes form.

Gr 7 · Unit 2 · #23 · 3.2.6-8.O

✂ cut on dashes · fold on dots

GR 7 · UNIT 2 STEELS 3.2.6-8.O #24

How can you show energy before and after a transfer?

Make an energy inventory: a list or diagram of where energy is before and after, using evidence like motion or temperature changes.

Gr 7 · Unit 2 · #24 · 3.2.6-8.O

✂ cut on dashes · fold on dots

GR 7 · UNIT 2 STEELS 3.2.6-8.P #25

Potential energy

Stored energy in a system that depends on the positions of the objects in it. Example: a book on a high shelf.

Gr 7 · Unit 2 · #25 · 3.2.6-8.P

✂ cut on dashes · fold on dots

GR 7 · UNIT 2 STEELS 3.2.6-8.P #26

How does height change stored energy with Earth?

The higher an object is above the ground, the more potential energy the object-Earth system stores. A roller coaster cart has the most at the top of the hill.

Gr 7 · Unit 2 · #26 · 3.2.6-8.P

✂ cut on dashes · fold on dots

GR 7 · UNIT 2 STEELS 3.2.6-8.P #27

Can magnets store potential energy?

Yes. Changing the distance between two magnets, or turning one around, changes how much potential energy is stored.

Gr 7 · Unit 2 · #27 · 3.2.6-8.P

✂ cut on dashes · fold on dots

GR 7 · UNIT 2 STEELS 3.2.6-8.P #28

What changes when a charged balloon nears someone's hair?

As the distance changes, the amount of stored electric potential energy in the balloon-hair system changes.

Gr 7 · Unit 2 · #28 · 3.2.6-8.P

✂ cut on dashes · fold on dots

GR 7 · UNIT 2 STEELS 3.2.6-8.P #29

How are forces and energy transfer connected?

When two objects interact, each pushes or pulls on the other. These forces can transfer energy to or from the objects.

Gr 7 · Unit 2 · #29 · 3.2.6-8.P

✂ cut on dashes · fold on dots

Grade 7 · Unit 3 · Waves and Information — Flashcards

PA STEELS 3.2.6-8.Q, R, S

GR 7 · UNIT 3 | STEELS 3.2.6-8.Q | #1

Wave

A repeating disturbance that carries energy from place to place. Examples: ripples on water or sound.

Gr 7 · Unit 3 · #1 · 3.2.6-8.Q

✂ cut on dashes · fold on dots

GR 7 · UNIT 3 | STEELS 3.2.6-8.Q | #2

Wavelength

The distance from one point on a wave to the same point on the next wave, such as top to top.

Gr 7 · Unit 3 · #2 · 3.2.6-8.Q

✂ cut on dashes · fold on dots

GR 7 · UNIT 3 | STEELS 3.2.6-8.Q | #3

Frequency

How many waves pass a point each second. More waves per second means a higher frequency.

Gr 7 · Unit 3 · #3 · 3.2.6-8.Q

✂ cut on dashes · fold on dots

GR 7 · UNIT 3 STEELS 3.2.6-8.Q #4

Amplitude

How big a wave is: the distance from its resting position to its highest point.

Gr 7 · Unit 3 · #4 · 3.2.6-8.Q

✂ cut on dashes · fold on dots

GR 7 · UNIT 3 STEELS 3.2.6-8.Q #5

How is a wave's amplitude related to its energy?

A wave with a larger amplitude carries more energy. A big ocean wave hits harder than a small ripple.

Gr 7 · Unit 3 · #5 · 3.2.6-8.Q

✂ cut on dashes · fold on dots

GR 7 · UNIT 3 STEELS 3.2.6-8.R #6

Mechanical wave

A wave that travels through matter, like sound through air or a wave along a rope.

Gr 7 · Unit 3 · #6 · 3.2.6-8.R

✂ cut on dashes · fold on dots

GR 7 · UNIT 3 STEELS 3.2.6-8.R #7

Medium

The matter a wave travels through, such as air, water, or a solid. Sound needs a medium.

Gr 7 · Unit 3 · #7 · 3.2.6-8.R

✂ cut on dashes · fold on dots

GR 7 · UNIT 3 STEELS 3.2.6-8.R #8

Can sound travel through empty space?

No. A sound wave needs a medium, like air or water, to travel through.

Gr 7 · Unit 3 · #8 · 3.2.6-8.R

✂ cut on dashes · fold on dots

GR 7 · UNIT 3 STEELS 3.2.6-8.R #9

Reflection

When a wave bounces off a surface. Example: light off a mirror or an echo off a wall.

Gr 7 · Unit 3 · #9 · 3.2.6-8.R

✂ cut on dashes · fold on dots

GR 7 · UNIT 3 STEELS 3.2.6-8.R #10

Absorption

When a material takes in a wave's energy. Example: a dark shirt absorbs light and warms up.

Gr 7 · Unit 3 · #10 · 3.2.6-8.R

✂ cut on dashes · fold on dots

GR 7 · UNIT 3 STEELS 3.2.6-8.R #11

Transmission

When a wave passes through a material. Example: light passing through a window.

Gr 7 · Unit 3 · #11 · 3.2.6-8.R

✂ cut on dashes · fold on dots

GR 7 · UNIT 3 STEELS 3.2.6-8.R #12

What decides if light is reflected, absorbed, or transmitted?

The object's material and the frequency (color) of the light.

Gr 7 · Unit 3 · #12 · 3.2.6-8.R

✂ cut on dashes · fold on dots

GR 7 · UNIT 3 STEELS 3.2.6-8.R #13

What does the color of light tell you about it?

Color is tied to frequency. Different colors of light are waves with different frequencies.

Gr 7 · Unit 3 · #13 · 3.2.6-8.R

✂ cut on dashes · fold on dots

GR 7 · UNIT 3 STEELS 3.2.6-8.R #14

Transparent

Describes a material that lets light pass through it so you can see through it, like air, water, or glass.

Gr 7 · Unit 3 · #14 · 3.2.6-8.R

✂ cut on dashes · fold on dots

GR 7 · UNIT 3 STEELS 3.2.6-8.R #15

What path does light travel?

Light travels in straight lines, except where it crosses from one transparent material to another.

Gr 7 · Unit 3 · #15 · 3.2.6-8.R

✂ cut on dashes · fold on dots

GR 7 · UNIT 3 STEELS 3.2.6-8.R #16

What happens to light going from air into water?

Its path bends at the surface between the two materials. That is why a straw in water looks bent.

Gr 7 · Unit 3 · #16 · 3.2.6-8.R

✂ cut on dashes · fold on dots

GR 7 · UNIT 3 STEELS 3.2.6-8.R #17

What does a wave model of light help explain?

Brightness, color, and how light bends by different amounts at a surface depending on its frequency.

Gr 7 · Unit 3 · #17 · 3.2.6-8.R

✂ cut on dashes · fold on dots

GR 7 · UNIT 3 STEELS 3.2.6-8.R #18

Why can't light be a matter wave like sound?

Light can travel through empty space, where there is no matter. Sound and water waves need matter.

Gr 7 · Unit 3 · #18 · 3.2.6-8.R

✂ cut on dashes · fold on dots

GR 7 · UNIT 3 STEELS 3.2.6-8.S #19

Signal

A wave or pattern used to carry information from one place to another, such as radio waves or light pulses.

Gr 7 · Unit 3 · #19 · 3.2.6-8.S

✂ cut on dashes · fold on dots

GR 7 · UNIT 3 STEELS 3.2.6-8.S #20

Encode

To change information into a form that can be sent or stored, like turning a voice into a signal.

Gr 7 · Unit 3 · #20 · 3.2.6-8.S

✂ cut on dashes · fold on dots

GR 7 · UNIT 3 STEELS 3.2.6-8.S #21

Transmit

To send a signal from one place to another, like through a cable or through the air.

Gr 7 · Unit 3 · #21 · 3.2.6-8.S

✂ cut on dashes · fold on dots

GR 7 · UNIT 3 STEELS 3.2.6-8.S #22

Analog signal

A signal that changes smoothly and continuously to copy the original information, like a wave shaped like a sound.

Gr 7 · Unit 3 · #22 · 3.2.6-8.S

✂ cut on dashes · fold on dots

GR 7 · UNIT 3 STEELS 3.2.6-8.S #23

Digital signal

A signal sent as separate wave pulses, like on/off patterns of 0s and 1s. Also called a digitized signal.

Gr 7 · Unit 3 · #23 · 3.2.6-8.S

✂ cut on dashes · fold on dots

GR 7 · UNIT 3 STEELS 3.2.6-8.S #24

Wave pulse

A single short burst of a wave. Digital signals are sent as series of wave pulses.

Gr 7 · Unit 3 · #24 · 3.2.6-8.S

✂ cut on dashes · fold on dots

GR 7 · UNIT 3 STEELS 3.2.6-8.S #25

Binary pattern

Information stored as a pattern of only two values, 0 and 1. Computers turn these patterns into sound or text.

Gr 7 · Unit 3 · #25 · 3.2.6-8.S

✂ cut on dashes · fold on dots

GR 7 · UNIT 3 STEELS 3.2.6-8.S #26

Why are digital signals more reliable than analog?

Each pulse only has to be read as on or off, so small disturbances don't change the message. Analog signals get changed more easily.

Gr 7 · Unit 3 · #26 · 3.2.6-8.S

✂ cut on dashes · fold on dots

GR 7 · UNIT 3 STEELS 3.2.6-8.S #27

Fiber optic cable

A thin glass or plastic cable that carries information as pulses of light.

Gr 7 · Unit 3 · #27 · 3.2.6-8.S

✂ cut on dashes · fold on dots

GR 7 · UNIT 3 STEELS 3.2.6-8.S #28

How does Wi-Fi send information?

Wi-Fi devices send information as pulses of radio waves through the air.

Gr 7 · Unit 3 · #28 · 3.2.6-8.S

✂ cut on dashes · fold on dots

GR 7 · UNIT 4 STEELS 3.3.6-8.A #1

Model (in science)

A way to show how something works when it is too big, small, or far to study directly. Models can be physical, graphical (drawings), or conceptual (ideas).

Gr 7 · Unit 4 · #1 · 3.3.6-8.A

✂ cut on dashes · fold on dots

GR 7 · UNIT 4 STEELS 3.3.6-8.A #2

Earth-sun-moon system

The sun, Earth, and moon and how they move: the moon orbits Earth while Earth orbits the sun. Its motions cause moon phases, eclipses, and seasons.

Gr 7 · Unit 4 · #2 · 3.3.6-8.A

✂ cut on dashes · fold on dots

GR 7 · UNIT 4 STEELS 3.3.6-8.A #3

Orbit

The curved path one object takes around another in space. Example: Earth orbits the sun once a year, and the moon orbits Earth.

Gr 7 · Unit 4 · #3 · 3.3.6-8.A

✂ cut on dashes · fold on dots

GR 7 · UNIT 4 STEELS 3.3.6-8.A #4

Earth's spin axis

An imaginary line through the North and South Poles that Earth spins around. It is tilted compared to Earth's orbit and points the same direction all year.

Gr 7 · Unit 4 · #4 · 3.3.6-8.A

✂ cut on dashes · fold on dots

GR 7 · UNIT 4 STEELS 3.3.6-8.A #5

Why do the sun and stars seem to move across the sky?

Because Earth spins. As Earth turns, the sun, moon, and stars appear to rise and set in patterns we can observe and predict.

Gr 7 · Unit 4 · #5 · 3.3.6-8.A

✂ cut on dashes · fold on dots

GR 7 · UNIT 4 STEELS 3.3.6-8.A #6

Cyclic pattern

A pattern that repeats over and over in the same order. Moon phases, eclipses, and seasons are cyclic patterns caused by motions in the Earth-sun-moon system.

Gr 7 · Unit 4 · #6 · 3.3.6-8.A

✂ cut on dashes · fold on dots

GR 7 · UNIT 4 STEELS 3.3.6-8.A #7

Lunar phases

The changing shapes of the moon we see. The sun always lights half the moon; as the moon orbits Earth, we see different amounts of that lit half.

Gr 7 · Unit 4 · #7 · 3.3.6-8.A

✂ cut on dashes · fold on dots

GR 7 · UNIT 4 STEELS 3.3.6-8.A #8

Eclipse

When one object in space blocks sunlight from reaching another, casting a shadow. Earth and the moon can both be part of eclipses.

Gr 7 · Unit 4 · #8 · 3.3.6-8.A

✂ cut on dashes · fold on dots

GR 7 · UNIT 4 STEELS 3.3.6-8.A #9

Solar eclipse

An eclipse of the sun. The moon passes between the sun and Earth, and the moon's shadow falls on Earth, blocking our view of the sun.

Gr 7 · Unit 4 · #9 · 3.3.6-8.A

✂ cut on dashes · fold on dots

GR 7 · UNIT 4 STEELS 3.3.6-8.A #10

Lunar eclipse

An eclipse of the moon. Earth passes between the sun and the moon, and Earth's shadow falls on the moon, making it look dark or reddish.

Gr 7 · Unit 4 · #10 · 3.3.6-8.A

✂ cut on dashes · fold on dots

GR 7 · UNIT 4 STEELS 3.3.6-8.A #11

Solar eclipse vs. lunar eclipse

Solar eclipse: the moon is in the middle, and its shadow hits Earth. Lunar eclipse: Earth is in the middle, and its shadow hits the moon.

Gr 7 · Unit 4 · #11 · 3.3.6-8.A

✂ cut on dashes · fold on dots

GR 7 · UNIT 4 STEELS 3.3.6-8.A #12

Seasons

The yearly cycle of spring, summer, fall, and winter. They happen because Earth's spin axis is tilted as Earth orbits the sun.

Gr 7 · Unit 4 · #12 · 3.3.6-8.A

✂ cut on dashes · fold on dots

GR 7 · UNIT 4 | STEELS 3.3.6-8.A | #13

Why is it warmer in summer than in winter?

Because of Earth's tilt, sunlight hits our part of Earth more directly in summer, so it is more intense. In winter, the same sunlight is spread out and weaker.

Gr 7 · Unit 4 · #13 · 3.3.6-8.A

✂ cut on dashes · fold on dots

GR 7 · UNIT 4 | STEELS 3.3.6-8.B | #14

Gravity

A force of attraction that pulls objects toward each other. Gravity holds the solar system and the Milky Way galaxy together and controls orbits.

Gr 7 · Unit 4 · #14 · 3.3.6-8.B

✂ cut on dashes · fold on dots

GR 7 · UNIT 4 | STEELS 3.3.6-8.B | #15

Solar system

The sun and all the objects held in orbit around it by its gravity, including planets, their moons, and asteroids.

Gr 7 · Unit 4 · #15 · 3.3.6-8.B

✂ cut on dashes · fold on dots

GR 7 · UNIT 4 | STEELS 3.3.6-8.B | #16

Sun

The star at the center of our solar system. Its gravity pulls on the planets and other objects and keeps them in orbit.

Gr 7 · Unit 4 · #16 · 3.3.6-8.B

✂ cut on dashes · fold on dots

GR 7 · UNIT 4 | STEELS 3.3.6-8.B | #17

Planet

A large object that orbits the sun. Earth is one of the planets in our solar system.

Gr 7 · Unit 4 · #17 · 3.3.6-8.B

✂ cut on dashes · fold on dots

GR 7 · UNIT 4 | STEELS 3.3.6-8.B | #18

Moon

A natural object that orbits a planet. Earth has one moon; some planets have many moons.

Gr 7 · Unit 4 · #18 · 3.3.6-8.B

✂ cut on dashes · fold on dots

GR 7 · UNIT 4 | STEELS 3.3.6-8.B | #19

Asteroid

A rocky object, smaller than a planet, that orbits the sun.

Gr 7 · Unit 4 · #19 · 3.3.6-8.B

✂ cut on dashes · fold on dots

GR 7 · UNIT 4 | STEELS 3.3.6-8.B | #20

What keeps the planets in orbit around the sun?

The sun's gravity. It constantly pulls on the planets, bending their paths into orbits so they don't fly off into space.

Gr 7 · Unit 4 · #20 · 3.3.6-8.B

✂ cut on dashes · fold on dots

GR 7 · UNIT 4 STEELS 3.3.6-8.B #21

Elliptical orbit

An orbit shaped like a stretched circle (an oval). The orbits of planets around the sun are elliptical, though many are very close to circles.

Gr 7 · Unit 4 · #21 · 3.3.6-8.B

✂ cut on dashes · fold on dots

GR 7 · UNIT 4 STEELS 3.3.6-8.B #22

How did the solar system form?

It appears to have formed from a spinning disk of dust and gas. Gravity pulled the material together to form the sun, planets, and other objects.

Gr 7 · Unit 4 · #22 · 3.3.6-8.B

✂ cut on dashes · fold on dots

GR 7 · UNIT 4 STEELS 3.3.6-8.B #23

Galaxy

A huge group of stars, plus gas and dust, held together by gravity.

Gr 7 · Unit 4 · #23 · 3.3.6-8.B

✂ cut on dashes · fold on dots

GR 7 · UNIT 4 STEELS 3.3.6-8.B #24

Milky Way galaxy

The galaxy that our solar system is part of. Gravity holds its stars together, and our sun is just one of them.

Gr 7 · Unit 4 · #24 · 3.3.6-8.B

✂ cut on dashes · fold on dots

GR 7 · UNIT 4 STEELS 3.3.6-8.B #25

Universe

Everything that exists in space, including all the galaxies. The Milky Way is one of many galaxies in the universe.

Gr 7 · Unit 4 · #25 · 3.3.6-8.B

✂ cut on dashes · fold on dots

GR 7 · UNIT 4 STEELS 3.3.6-8.C #26

Scale

How big or far something is compared to something else. Example: if the sun were a basketball, Earth would be smaller than a peppercorn.

Gr 7 · Unit 4 · #26 · 3.3.6-8.C

✂ cut on dashes · fold on dots

GR 7 · UNIT 4 STEELS 3.3.6-8.C #27

Scale model

A model that keeps the same size or distance ratios as the real thing, just smaller. Example: showing planet distances along a football field.

Gr 7 · Unit 4 · #27 · 3.3.6-8.C

✂ cut on dashes · fold on dots

GR 7 · UNIT 4 STEELS 3.3.6-8.C #28

Scale properties

Measurable features of an object in space, such as its size, the thickness of its layers, the size of surface features, and its orbital radius.

Gr 7 · Unit 4 · #28 · 3.3.6-8.C

✂ cut on dashes · fold on dots

GR 7 · UNIT 4 STEELS 3.3.6-8.C #29

Orbital radius

The distance from an object to the object it orbits. Example: Earth's orbital radius is its distance from the sun.

Gr 7 · Unit 4 · #29 · 3.3.6-8.C

✂ cut on dashes · fold on dots

GR 7 · UNIT 4 STEELS 3.3.6-8.C #30

Crust

The thin, solid, rocky outer layer of a rocky planet or moon, like Earth or Mars. Scientists can compare the size of this layer on different objects.

Gr 7 · Unit 4 · #30 · 3.3.6-8.C

✂ cut on dashes · fold on dots

GR 7 · UNIT 4 STEELS 3.3.6-8.C #31

Surface feature

A landform on the outside of a planet or moon, such as a volcano. Scientists compare the sizes of surface features on different objects.

Gr 7 · Unit 4 · #31 · 3.3.6-8.C

✂ cut on dashes · fold on dots

GR 7 · UNIT 4 STEELS 3.3.6-8.C #32

How do scientists get data about objects in the solar system?

From instruments on Earth, telescopes in space, and spacecraft. The data can be numbers, drawings, photographs, and models.

Gr 7 · Unit 4 · #32 · 3.3.6-8.C

✂ cut on dashes · fold on dots

GR 7 · UNIT 4 STEELS 3.3.6-8.C #33

Space-based telescope

A telescope that orbits in space, above Earth's atmosphere, so it can collect clearer data about distant objects.

Gr 7 · Unit 4 · #33 · 3.3.6-8.C

✂ cut on dashes · fold on dots

GR 7 · UNIT 4 STEELS 3.3.6-8.C #34

Spacecraft

A vehicle sent into space. Spacecraft can travel to other planets and moons to take photos and measurements up close.

Gr 7 · Unit 4 · #34 · 3.3.6-8.C

✂ cut on dashes · fold on dots

GR 7 · UNIT 4 STEELS 3.3.6-8.C #35

Why compare data about different solar system objects?

To find how they are alike and different, such as which planet has thicker layers, bigger volcanoes, or a larger orbit.

Gr 7 · Unit 4 · #35 · 3.3.6-8.C

✂ cut on dashes · fold on dots

Grade 7 · Unit 5 · Earth's Resources and Human Impact — Flashcards

PA STEELS 3.3.6-8.K, M, N

GR 7 · UNIT 5 | STEELS 3.3.6-8.K | #1

Natural resource

Something from nature that people use, from Earth's land, ocean, atmosphere, or living things. Examples: minerals, fresh water, soil, and forests.

Gr 7 · Unit 5 · #1 · 3.3.6-8.K

✂ cut on dashes · fold on dots

GR 7 · UNIT 5 | STEELS 3.3.6-8.K | #2

Biosphere

All the living things on Earth and the places where they live. People get biosphere resources from it, such as food and wood.

Gr 7 · Unit 5 · #2 · 3.3.6-8.K

✂ cut on dashes · fold on dots

GR 7 · UNIT 5 | STEELS 3.3.6-8.K | #3

Nonrenewable resource

A natural resource that cannot be replaced within a human lifetime once it is used up. Examples: petroleum and metal ores.

Gr 7 · Unit 5 · #3 · 3.3.6-8.K

✂ cut on dashes · fold on dots

GR 7 · UNIT 5 STEELS 3.3.6-8.K #4

Renewable resource

A natural resource that can be replaced by nature within a human lifetime if it is used carefully. Example: trees that are replanted.

Gr 7 · Unit 5 · #4 · 3.3.6-8.K

✂ cut on dashes · fold on dots

GR 7 · UNIT 5 STEELS 3.3.6-8.K #5

Why are many natural resources limited?

Earth has only so much of them, and many form far slower than people use them. Minerals, fresh water, and biosphere resources are all limited.

Gr 7 · Unit 5 · #5 · 3.3.6-8.K

✂ cut on dashes · fold on dots

GR 7 · UNIT 5 STEELS 3.3.6-8.K #6

Energy resource

A natural resource people use to get energy. Example: petroleum (oil) found underground.

Gr 7 · Unit 5 · #6 · 3.3.6-8.K

✂ cut on dashes · fold on dots

GR 7 · UNIT 5 STEELS 3.3.6-8.K #7

Groundwater

Fresh water stored underground in the spaces in soil and rock. People pump it up from wells to use.

Gr 7 · Unit 5 · #7 · 3.3.6-8.K

✂ cut on dashes · fold on dots

GR 7 · UNIT 5 STEELS 3.3.6-8.K #8

Aquifer

An underground layer of rock or sediment that holds groundwater. People withdraw water from aquifers using wells.

Gr 7 · Unit 5 · #8 · 3.3.6-8.K

✂ cut on dashes · fold on dots

GR 7 · UNIT 5 STEELS 3.3.6-8.K #9

Why are Earth's resources found in some places but not others?

Because of past and current geologic processes. Resources form only where the right conditions happened, so they are spread unevenly around Earth.

Gr 7 · Unit 5 · #9 · 3.3.6-8.K

✂ cut on dashes · fold on dots

GR 7 · UNIT 5 STEELS 3.3.6-8.K #10

Petroleum

Oil formed from the remains of ocean life buried in sediment long ago. It is found where rock layers trapped it underground.

Gr 7 · Unit 5 · #10 · 3.3.6-8.K

✂ cut on dashes · fold on dots

GR 7 · UNIT 5 STEELS 3.3.6-8.K #11

Metal ore

Rock that contains enough metal to be worth mining. Many ores are found where hot water and volcanoes were active long ago near subduction zones.

Gr 7 · Unit 5 · #11 · 3.3.6-8.K

✂ cut on dashes · fold on dots

GR 7 · UNIT 5 STEELS 3.3.6-8.K #12

Subduction zone

A place where one tectonic plate sinks under another. Volcanoes and hot water there can leave behind metal ores in the rock.

Gr 7 · Unit 5 · #12 · 3.3.6-8.K

✂ cut on dashes · fold on dots

GR 7 · UNIT 5 STEELS 3.3.6-8.K #13

Soil

A mix of weathered rock bits, minerals, and decayed living matter. It forms where rock is actively weathered or where sediment is deposited.

Gr 7 · Unit 5 · #13 · 3.3.6-8.K

✂ cut on dashes · fold on dots

GR 7 · UNIT 5 STEELS 3.3.6-8.K #14

How are humans changing where resources are found?

By removing them. As people mine, drill, and pump resources, the amounts left and where they are found change a lot.

Gr 7 · Unit 5 · #14 · 3.3.6-8.K

✂ cut on dashes · fold on dots

GR 7 · UNIT 5 STEELS 3.3.6-8.M #15

Human impact

A change that people cause in the environment. Examples: using water, building on land, and causing pollution.

Gr 7 · Unit 5 · #15 · 3.3.6-8.M

✂ cut on dashes · fold on dots

GR 7 · UNIT 5 STEELS 3.3.6-8.M #16

Habitat

The natural home of a living thing, where it finds food, water, and shelter. Human activities can damage or destroy habitats.

Gr 7 · Unit 5 · #16 · 3.3.6-8.M

✂ cut on dashes · fold on dots

GR 7 · UNIT 5 STEELS 3.3.6-8.M #17

Extinction

When every member of a species has died out. Human activities, such as destroying habitats, have caused the extinction of some species.

Gr 7 · Unit 5 · #17 · 3.3.6-8.M

✂ cut on dashes · fold on dots

GR 7 · UNIT 5 STEELS 3.3.6-8.M #18

Pollution

Harmful materials added to the air, water, or land. Example: smoke from factories polluting the air.

Gr 7 · Unit 5 · #18 · 3.3.6-8.M

✂ cut on dashes · fold on dots

GR 7 · UNIT 5 STEELS 3.3.6-8.M #19

How can people's water use affect the environment?

Taking water from streams and aquifers, or building dams and levees, changes how water flows and how much is left for living things.

Gr 7 · Unit 5 · #19 · 3.3.6-8.M

✂ cut on dashes · fold on dots

GR 7 · UNIT 5 | STEELS 3.3.6-8.M | #20

Dam

A wall built across a river or stream to hold back water. Dams store water but change the river and its habitats.

Gr 7 · Unit 5 · #20 · 3.3.6-8.M

✂ cut on dashes · fold on dots

GR 7 · UNIT 5 | STEELS 3.3.6-8.M | #21

Levee

A raised wall of earth built along a river to keep it from flooding nearby land.

Gr 7 · Unit 5 · #21 · 3.3.6-8.M

✂ cut on dashes · fold on dots

GR 7 · UNIT 5 | STEELS 3.3.6-8.M | #22

How can people's land use affect the environment?

Building cities, farming, and removing wetlands replace natural habitats and change how water and soil move.

Gr 7 · Unit 5 · #22 · 3.3.6-8.M

✂ cut on dashes · fold on dots

GR 7 · UNIT 5 | STEELS 3.3.6-8.M | #23

Urban development

Building and growing cities and towns, with roads, buildings, and parking lots. It is one kind of human land use.

Gr 7 · Unit 5 · #23 · 3.3.6-8.M

✂ cut on dashes · fold on dots

GR 7 · UNIT 5 STEELS 3.3.6-8.M #24

Are changes to the environment always bad for living things?

No. A change can hurt some living things and help others. Example: a new lake from a dam helps some fish but floods land animals' homes.

Gr 7 · Unit 5 · #24 · 3.3.6-8.M

✂ cut on dashes · fold on dots

GR 7 · UNIT 5 STEELS 3.3.6-8.M #25

How can people monitor and reduce their impact on the environment?

Examine the impact, measure it over time, then design and test solutions that could reduce it, checking which ones are possible to do.

Gr 7 · Unit 5 · #25 · 3.3.6-8.M

✂ cut on dashes · fold on dots

GR 7 · UNIT 5 STEELS 3.3.6-8.N #26

Human population growth

An increase in the number of people living on Earth or in an area. More people usually means more natural resources are used.

Gr 7 · Unit 5 · #26 · 3.3.6-8.N

✂ cut on dashes · fold on dots

GR 7 · UNIT 5 STEELS 3.3.6-8.N #27

Consumption (of natural resources)

Using up natural resources, such as fresh water, minerals, energy, and food.

Gr 7 · Unit 5 · #27 · 3.3.6-8.N

✂ cut on dashes · fold on dots

GR 7 · UNIT 5 STEELS 3.3.6-8.N #28

Per capita consumption

The amount of a resource used per person. Example: total water used in a town divided by the number of people who live there.

Gr 7 · Unit 5 · #28 · 3.3.6-8.N

✂ cut on dashes · fold on dots

GR 7 · UNIT 5 STEELS 3.3.6-8.N #29

What happens when population and per capita consumption go up?

Negative impacts on Earth usually increase, unless the activities and technologies involved are engineered to reduce them.

Gr 7 · Unit 5 · #29 · 3.3.6-8.N

✂ cut on dashes · fold on dots

GR 7 · UNIT 5 STEELS 3.3.6-8.N #30

How can more people using more resources change Earth's systems?

It can change how Earth's systems look, what they are made of, and how their parts are arranged, and it can speed up how fast they change.

Gr 7 · Unit 5 · #30 · 3.3.6-8.N

✂ cut on dashes · fold on dots

GR 7 · UNIT 5 STEELS 3.3.6-8.N #31

Does science decide what society should do about human impacts?

No. Science describes the effects of more people and more resource use, but society makes the decisions about what actions to take.

Gr 7 · Unit 5 · #31 · 3.3.6-8.N

✂ cut on dashes · fold on dots

Grade 7 · Unit 6 · Environmental Literacy and Sustainability — Flashcards

PA STEELS 3.4.6-8.A, B, D, E, F, G, H, I

GR 7 · UNIT 6 STEELS 3.4.6-8.A #1

Agriculture

Farming: growing crops and raising animals for food, fiber, and energy. It is also a major way people use land.

Gr 7 · Unit 6 · #1 · 3.4.6-8.A

✂ cut on dashes · fold on dots

GR 7 · UNIT 6 STEELS 3.4.6-8.A #2

Food system

All the steps that get food to people: producing it on farms, processing it, moving and selling it, eating it, and handling the waste.

Gr 7 · Unit 6 · #2 · 3.4.6-8.A

✂ cut on dashes · fold on dots

GR 7 · UNIT 6 STEELS 3.4.6-8.A #3

Fiber (agricultural product)

Plant or animal material made into products like cloth. Examples: cotton from plants and wool from sheep.

Gr 7 · Unit 6 · #3 · 3.4.6-8.A

✂ cut on dashes · fold on dots

GR 7 · UNIT 6 STEELS 3.4.6-8.A #4

Processing (food and fiber)

Changing raw farm products into things people use. Examples: grinding wheat into flour or spinning cotton into thread.

Gr 7 · Unit 6 · #4 · 3.4.6-8.A

✂ cut on dashes · fold on dots

GR 7 · UNIT 6 STEELS 3.4.6-8.A #5

Inputs (in a system)

What goes into a system. On a farm, inputs include water, soil, energy, seeds, and fertilizer.

Gr 7 · Unit 6 · #5 · 3.4.6-8.A

✂ cut on dashes · fold on dots

GR 7 · UNIT 6 STEELS 3.4.6-8.A #6

Outputs (in a system)

What comes out of a system. On a farm, outputs include food, fiber, and waste.

Gr 7 · Unit 6 · #6 · 3.4.6-8.A

✂ cut on dashes · fold on dots

GR 7 · UNIT 6 STEELS 3.4.6-8.A #7

Life cycle of a product

All the stages of a product from start to finish. Example for a T-shirt: grow cotton, make cloth, sew, use it, then throw away or recycle.

Gr 7 · Unit 6 · #7 · 3.4.6-8.A

✂ cut on dashes · fold on dots

GR 7 · UNIT 6 STEELS 3.4.6-8.A #8

Sustainability

Using natural resources in a way that meets people's needs now while leaving enough for the future.

Gr 7 · Unit 6 · #8 · 3.4.6-8.A

✂ cut on dashes · fold on dots

GR 7 · UNIT 6 STEELS 3.4.6-8.B #9

Society

A large group of people who live together and share economic and social systems. Example: a country.

Gr 7 · Unit 6 · #9 · 3.4.6-8.B

✂ cut on dashes · fold on dots

GR 7 · UNIT 6 STEELS 3.4.6-8.B #10

Culture

The shared beliefs, customs, and ways of life of a group of people. Culture can affect how people use and care for natural resources.

Gr 7 · Unit 6 · #10 · 3.4.6-8.B

✂ cut on dashes · fold on dots

GR 7 · UNIT 6 STEELS 3.4.6-8.B #11

Economic system

The way a society makes, buys, sells, and shares goods and services. It affects how natural resources are used.

Gr 7 · Unit 6 · #11 · 3.4.6-8.B

✂ cut on dashes · fold on dots

GR 7 · UNIT 6 STEELS 3.4.6-8.B #12

Social system

The way people in a society are organized and interact, including their rules, groups, and traditions.

Gr 7 · Unit 6 · #12 · 3.4.6-8.B

✂ cut on dashes · fold on dots

GR 7 · UNIT 6 STEELS 3.4.6-8.B #13

Recycling

Collecting used materials, like paper, metal, and plastic, so they can be made into new products instead of thrown away.

Gr 7 · Unit 6 · #13 · 3.4.6-8.B

✂ cut on dashes · fold on dots

GR 7 · UNIT 6 STEELS 3.4.6-8.B #14

Waste management

How a community collects, moves, and gets rid of or reuses its trash, such as by landfills or recycling.

Gr 7 · Unit 6 · #14 · 3.4.6-8.B

✂ cut on dashes · fold on dots

GR 7 · UNIT 6 STEELS 3.4.6-8.B #15

Land development

Changing natural land for human use, such as building homes, roads, or businesses.

Gr 7 · Unit 6 · #15 · 3.4.6-8.B

✂ cut on dashes · fold on dots

GR 7 · UNIT 6 STEELS 3.4.6-8.B #16

Why do societies use and manage natural resources differently?

Their economic and social systems and cultures differ. Comparing data from two or more societies shows differences in farming, recycling, or fuel use.

Gr 7 · Unit 6 · #16 · 3.4.6-8.B

✂ cut on dashes · fold on dots

GR 7 · UNIT 6 STEELS 3.4.6-8.D #17

Sediment load

The amount of soil and sediment carried by a stream or river. Too much makes water muddy and can bury habitats, a problem in many PA waterways.

Gr 7 · Unit 6 · #17 · 3.4.6-8.D

✂ cut on dashes · fold on dots

GR 7 · UNIT 6 STEELS 3.4.6-8.D #18

Nutrient load

The amount of nutrients, such as from fertilizer, washed into a waterway. Too many nutrients can harm the living things in streams and rivers.

Gr 7 · Unit 6 · #18 · 3.4.6-8.D

✂ cut on dashes · fold on dots

GR 7 · UNIT 6 STEELS 3.4.6-8.D #19

Air quality

How clean or polluted the air is, indoors or outdoors. Poor air quality can harm people's health and other living things.

Gr 7 · Unit 6 · #19 · 3.4.6-8.D

✂ cut on dashes · fold on dots

GR 7 · UNIT 6 STEELS 3.4.6-8.D #20

Urban heat island

A city area that is hotter than the land around it because pavement and buildings soak up heat and there are fewer trees.

Gr 7 · Unit 6 · #20 · 3.4.6-8.D

✂ cut on dashes · fold on dots

GR 7 · UNIT 6 STEELS 3.4.6-8.D #21

Why are ecosystems called dynamic?

Because they are always changing. Their characteristics can vary over time, and a disruption to one part can shift all of its populations.

Gr 7 · Unit 6 · #21 · 3.4.6-8.D

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GR 7 · UNIT 6 STEELS 3.4.6-8.D #22

How can a PA environmental issue affect people and nature?

It can affect both human and natural systems. Example: extra sediment in a stream harms fish and can make drinking water harder to clean.

Gr 7 · Unit 6 · #22 · 3.4.6-8.D

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GR 7 · UNIT 6 STEELS 3.4.6-8.D #23

Why gather information from multiple sources?

Different sources give different facts and views. Putting them together (synthesizing) gives a fuller, more reliable picture of an issue.

Gr 7 · Unit 6 · #23 · 3.4.6-8.D

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GR 7 · UNIT 6 STEELS 3.4.6-8.E #24

Environmental data

Measurements and observations about an environment. Examples: weather data, stream studies, air quality readings, and biodiversity counts.

Gr 7 · Unit 6 · #24 · 3.4.6-8.E

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GR 7 · UNIT 6 STEELS 3.4.6-8.E #25

Stream study

An investigation where students collect data at a stream, such as water temperature, clarity, and the living things found there.

Gr 7 · Unit 6 · #25 · 3.4.6-8.E

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GR 7 · UNIT 6 STEELS 3.4.6-8.E #26

Biodiversity assessment

A study that counts and identifies the different kinds of living things in an area to measure its biodiversity.

Gr 7 · Unit 6 · #26 · 3.4.6-8.E

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GR 7 · UNIT 6 STEELS 3.4.6-8.E #27

Why collect data to describe a local environment?

Measuring and observing an outdoor area lets you describe it accurately with evidence instead of guesses.

Gr 7 · Unit 6 · #27 · 3.4.6-8.E

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GR 7 · UNIT 6 STEELS 3.4.6-8.E #28

What do organisms depend on in their environment?

On interactions with other living things and with nonliving factors, such as water, sunlight, and temperature.

Gr 7 · Unit 6 · #28 · 3.4.6-8.E

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GR 7 · UNIT 6 STEELS 3.4.6-8.E #29

What influences the weather and climate of a place?

Interactions among sunlight, ocean, atmosphere, ice, landforms, and living things. Latitude, altitude, and local geography also matter.

Gr 7 · Unit 6 · #29 · 3.4.6-8.E

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GR 7 · UNIT 6 STEELS 3.4.6-8.F #30

Pest

A living thing that causes problems for people, crops, or buildings. Examples: insects that eat crops or mice in a school.

Gr 7 · Unit 6 · #30 · 3.4.6-8.F

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GR 7 · UNIT 6 STEELS 3.4.6-8.F #31

Invasive species

A species brought to a new area where it spreads quickly and causes harm, often crowding out native living things.

Gr 7 · Unit 6 · #31 · 3.4.6-8.F

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GR 7 · UNIT 6 STEELS 3.4.6-8.F #32

Integrated pest management (IPM)

Controlling pests by combining several methods (biological, cultural, mechanical, chemical) to protect people and the environment.

Gr 7 · Unit 6 · #32 · 3.4.6-8.F

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GR 7 · UNIT 6 STEELS 3.4.6-8.F #33

Biological methods (IPM)

Using living things or natural conditions to control pests. Examples: ladybugs that eat aphids, or managing indoor air quality so pests and mold can't thrive.

Gr 7 · Unit 6 · #33 · 3.4.6-8.F

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GR 7 · UNIT 6 STEELS 3.4.6-8.F #34

Cultural methods (IPM)

Changing how people grow crops to prevent pests. Examples: planting crops that resist local pests, or crop rotation.

Gr 7 · Unit 6 · #34 · 3.4.6-8.F

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GR 7 · UNIT 6 STEELS 3.4.6-8.F #35

Crop rotation

Planting different crops in the same field in different seasons or years. It breaks up pests' food supply.

Gr 7 · Unit 6 · #35 · 3.4.6-8.F

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GR 7 · UNIT 6 STEELS 3.4.6-8.F #36

Mechanical methods (IPM)

Physically removing or blocking pests. Example: trapping pests.

Gr 7 · Unit 6 · #36 · 3.4.6-8.F

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GR 7 · UNIT 6 STEELS 3.4.6-8.F #37

Chemical methods (IPM)

Using chemical products to control pests. Example: cleaning surfaces in schools.

Gr 7 · Unit 6 · #37 · 3.4.6-8.F

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GR 7 · UNIT 6 STEELS 3.4.6-8.F #38

How can IPM improve indoor and outdoor environments?

It controls pests and invasive species while keeping schools' surfaces and air clean and helping maintain biodiversity.

Gr 7 · Unit 6 · #38 · 3.4.6-8.F

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GR 7 · UNIT 6 STEELS 3.4.6-8.G #39

Best management practices (BMPs)

Proven ways of managing resources to protect the environment. They cover stormwater, forests, land use, wildlife, and waste.

Gr 7 · Unit 6 · #39 · 3.4.6-8.G

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GR 7 · UNIT 6 STEELS 3.4.6-8.G #40

Stormwater

Rain or melted snow that runs off roofs, roads, and parking lots instead of soaking into the ground. It can carry pollution into streams.

Gr 7 · Unit 6 · #40 · 3.4.6-8.G

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GR 7 · UNIT 6 STEELS 3.4.6-8.G #41

Stormwater management

Ways to slow down, soak up, or clean stormwater so it causes less flooding and pollution. Example: a rain garden.

Gr 7 · Unit 6 · #41 · 3.4.6-8.G

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GR 7 · UNIT 6 STEELS 3.4.6-8.G #42

Forest management

Caring for forests so they stay healthy and keep providing wood, habitats, and clean water. Example: replanting trees after cutting.

Gr 7 · Unit 6 · #42 · 3.4.6-8.G

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GR 7 · UNIT 6 STEELS 3.4.6-8.G #43

Wildlife management

Caring for wild animal populations and their habitats so they stay healthy. Example: setting hunting and fishing limits.

Gr 7 · Unit 6 · #43 · 3.4.6-8.G

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GR 7 · UNIT 6 STEELS 3.4.6-8.G #44

Environmental law

A rule made by a government to protect the environment, such as limits on pollution.

Gr 7 · Unit 6 · #44 · 3.4.6-8.G

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GR 7 · UNIT 6 STEELS 3.4.6-8.G #45

At what levels are environmental laws made?

International (between countries), federal (the whole U.S.), state (like Pennsylvania), and local (your town or county).

Gr 7 · Unit 6 · #45 · 3.4.6-8.G

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GR 7 · UNIT 6 STEELS 3.4.6-8.G #46

What are BMPs and environmental laws meant to achieve?

Environmental sustainability: protecting resources and habitats so people and other living things can keep using them in the future.

Gr 7 · Unit 6 · #46 · 3.4.6-8.G

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GR 7 · UNIT 6 STEELS 3.4.6-8.H #47

Environmental issue

A problem caused by changes to the environment that affects people or other living things. Example: litter polluting a local stream.

Gr 7 · Unit 6 · #47 · 3.4.6-8.H

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GR 7 · UNIT 6 STEELS 3.4.6-8.H #48

Stewardship

Taking responsibility for caring for the environment and its resources. People who do this are called stewards of the environment.

Gr 7 · Unit 6 · #48 · 3.4.6-8.H

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GR 7 · UNIT 6 STEELS 3.4.6-8.H #49

How can students act as stewards of the environment?

By designing and carrying out solutions to environmental issues. Examples: drawing a plan to plant trees, then doing the project.

Gr 7 · Unit 6 · #49 · 3.4.6-8.H

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GR 7 · UNIT 6 STEELS 3.4.6-8.H #50

How do you evaluate possible solutions to a problem?

Use a step-by-step process to check how well each solution meets the criteria and constraints, then choose the best one.

Gr 7 · Unit 6 · #50 · 3.4.6-8.H

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GR 7 · UNIT 6 STEELS 3.4.6-8.I #51

Environmental justice

The idea that all people, no matter their race, income, or where they live, deserve a healthy environment and fair protection from pollution.

Gr 7 · Unit 6 · #51 · 3.4.6-8.I

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GR 7 · UNIT 6 STEELS 3.4.6-8.I #52

Social equity

Fairness in how resources, opportunities, and harms are shared among people, such as clean air, clean water, and parks.

Gr 7 · Unit 6 · #52 · 3.4.6-8.I

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GR 7 · UNIT 6 STEELS 3.4.6-8.I #53

Systemic inequality

Unfairness built into a society's systems over time. It can leave some communities with more pollution or fewer parks than others.

Gr 7 · Unit 6 · #53 · 3.4.6-8.I

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GR 7 · UNIT 6 STEELS 3.4.6-8.I #54

Superfund site

A place so polluted by hazardous waste that the U.S. government (EPA) lists it for cleanup. It can harm the health of people living nearby.

Gr 7 · Unit 6 · #54 · 3.4.6-8.I

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GR 7 · UNIT 6 STEELS 3.4.6-8.I #55

Acid mine drainage

Acidic water that flows out of old mines, often carrying metals. It can turn streams orange and harm the living things in them; common in PA.

Gr 7 · Unit 6 · #55 · 3.4.6-8.I

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GR 7 · UNIT 6 STEELS 3.4.6-8.I #56

Green space

An area of grass, trees, or other plants in or near a community, such as a park. Not all neighborhoods have equal access to green space.

Gr 7 · Unit 6 · #56 · 3.4.6-8.I

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GR 7 · UNIT 6 STEELS 3.4.6-8.I #57

Water quality

How clean and safe water is for people and other living things to use.

Gr 7 · Unit 6 · #57 · 3.4.6-8.I

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GR 7 · UNIT 6 STEELS 3.4.6-8.I #58

How can environmental conditions affect environmental justice?

Some communities, now and in the past, have faced worse air, polluted water, or more heat, which harms health. That unfairness is an environmental justice issue.

Gr 7 · Unit 6 · #58 · 3.4.6-8.I

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GR 7 · UNIT 6 STEELS 3.4.6-8.I #59

How can changes in biodiversity affect people?

They can affect resources people need, like food, energy, and medicines, and ecosystem services such as cleaning and recycling water.

Gr 7 · Unit 6 · #59 · 3.4.6-8.I

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Grade 7 · Unit 7 · Engineering Design — Flashcards

PA STEELS 3.5.6-8.DD, II, JJ, M, N, O, P, Q, R, S, T, U, V, W, X

GR 7 · UNIT 7 | STEELS 3.5.6-8.Q | #1

Design thinking process

A step-by-step way to solve design problems: generate ideas, build possible solutions, test them, and redesign. Designers often go back and repeat steps.

Gr 7 · Unit 7 · #1 · 3.5.6-8.Q

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GR 7 · UNIT 7 | STEELS 3.5.6-8.R | #2

Problem finding

Spotting a need or want that a new product or system could solve. Ideas often come from problems in your own life or your family's lives.

Gr 7 · Unit 7 · #2 · 3.5.6-8.R

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GR 7 · UNIT 7 | STEELS 3.5.6-8.R | #3

Why do people develop new products and systems?

To solve problems and extend what people can do, based on the needs and wants of a person or a group. Example: a teen designed a filter to remove a livestock antibiotic from water.

Gr 7 · Unit 7 · #3 · 3.5.6-8.R

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GR 7 · UNIT 7 STEELS 3.5.6-8.W #4

Criteria

What a design solution must do to be successful. Example: a backpack must hold 5 books.

Gr 7 · Unit 7 · #4 · 3.5.6-8.W

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GR 7 · UNIT 7 STEELS 3.5.6-8.W #5

Constraints

Limits on a design solution, such as cost, time, materials, scientific principles, or effects on people and the environment.

Gr 7 · Unit 7 · #5 · 3.5.6-8.W

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GR 7 · UNIT 7 STEELS 3.5.6-8.W #6

Why define criteria and constraints precisely?

The more clearly a design task's criteria and constraints are defined, the more likely the solution will be successful.

Gr 7 · Unit 7 · #6 · 3.5.6-8.W

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GR 7 · UNIT 7 STEELS 3.5.6-8.W #7

What can limit the possible solutions to a design problem?

Scientific principles (like the laws of physics) and possible impacts on people and the natural environment.

Gr 7 · Unit 7 · #7 · 3.5.6-8.W

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GR 7 · UNIT 7 STEELS 3.5.6-8.II #8

Why predict outcomes at the start of a design?

Careful designers think about possible results of a product before it is finished, so they can avoid problems and make better choices early.

Gr 7 · Unit 7 · #8 · 3.5.6-8.II

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GR 7 · UNIT 7 STEELS 3.5.6-8.II #9

Ideation

Coming up with many ideas for solving a problem. It is an early part of the design process.

Gr 7 · Unit 7 · #9 · 3.5.6-8.II

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GR 7 · UNIT 7 STEELS 3.5.6-8.S #10

Design approach

The method or strategy a designer chooses to solve a problem. Different approaches have different benefits and opportunities.

Gr 7 · Unit 7 · #10 · 3.5.6-8.S

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GR 7 · UNIT 7 STEELS 3.5.6-8.S #11

How should a designer choose a final approach?

By carefully weighing the benefits and opportunities of each possible approach before picking one.

Gr 7 · Unit 7 · #11 · 3.5.6-8.S

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GR 7 · UNIT 7 STEELS 3.5.6-8.Q #12

Design fixation

Getting stuck on just one solution. Going back to earlier steps of the design process helps designers avoid it.

Gr 7 · Unit 7 · #12 · 3.5.6-8.Q

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GR 7 · UNIT 7 STEELS 3.5.6-8.T #13

Human factors

Things about people that affect how a design should work, such as how easy it is to use and how well it fits the body.

Gr 7 · Unit 7 · #13 · 3.5.6-8.T

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GR 7 · UNIT 7 STEELS 3.5.6-8.T #14

Ease of use

How simple a product is for people to understand and operate. Example: a can opener that works without much strength.

Gr 7 · Unit 7 · #14 · 3.5.6-8.T

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GR 7 · UNIT 7 STEELS 3.5.6-8.T #15

Ergonomics

Designing things to fit the human body so they are comfortable and safe. Example: a computer mouse shaped to fit your hand.

Gr 7 · Unit 7 · #15 · 3.5.6-8.T

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GR 7 · UNIT 7 STEELS 3.5.6-8.T #16

Unintended consequence

A result of a design that the designer did not plan or expect. It can be helpful or harmful.

Gr 7 · Unit 7 · #16 · 3.5.6-8.T

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GR 7 · UNIT 7 STEELS 3.5.6-8.M #17

How do models help engineers?

Models of all kinds, such as drawings, physical models, and computer models, let engineers test solutions and collect data.

Gr 7 · Unit 7 · #17 · 3.5.6-8.M

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GR 7 · UNIT 7 STEELS 3.5.6-8.M #18

Iterative testing

Testing a design, changing it based on the results, then testing again, over and over. Each round improves the design.

Gr 7 · Unit 7 · #18 · 3.5.6-8.M

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GR 7 · UNIT 7 STEELS 3.5.6-8.M #19

Optimal design

The best possible solution for the problem. It is reached by repeatedly testing the most promising designs and refining them.

Gr 7 · Unit 7 · #19 · 3.5.6-8.M

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GR 7 · UNIT 7 STEELS 3.5.6-8.V #20

Trade-off

A compromise in which one thing is given up to get something else that is wanted. Example: a stronger bridge may cost more.

Gr 7 · Unit 7 · #20 · 3.5.6-8.V

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GR 7 · UNIT 7 STEELS 3.5.6-8.V #21

Optimization

Making a design as good as it can be within its criteria and constraints. It usually requires trade-offs.

Gr 7 · Unit 7 · #21 · 3.5.6-8.V

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GR 7 · UNIT 7 STEELS 3.5.6-8.V #22

Refining a design

Making small improvements to a design so it better meets its criteria and constraints.

Gr 7 · Unit 7 · #22 · 3.5.6-8.V

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GR 7 · UNIT 7 STEELS 3.5.6-8.P #23

Competing design solutions

Different designs that try to solve the same problem. Engineers compare them to choose the best one.

Gr 7 · Unit 7 · #23 · 3.5.6-8.P

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GR 7 · UNIT 7 STEELS 3.5.6-8.P #24

How do engineers compare competing designs?

They use a systematic (step-by-step) process to check how well each design meets the same criteria and constraints.

Gr 7 · Unit 7 · #24 · 3.5.6-8.P

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GR 7 · UNIT 7 STEELS 3.5.6-8.N #25

What if no design wins every test?

Find which design did best on each test. Those best features can be used in the redesign.

Gr 7 · Unit 7 · #25 · 3.5.6-8.N

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GR 7 · UNIT 7 STEELS 3.5.6-8.N #26

Can parts of different designs be combined?

Yes. Combining the best parts of different solutions can create a new solution that is better than any of the earlier ones.

Gr 7 · Unit 7 · #26 · 3.5.6-8.N

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GR 7 · UNIT 7 STEELS 3.5.6-8.JJ #27

Redesign

Changing a design based on test results to make it work better.

Gr 7 · Unit 7 · #27 · 3.5.6-8.JJ

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GR 7 · UNIT 7 STEELS 3.5.6-8.JJ #28

What can creative problem-solving improve?

It can lead to greater efficiency, better performance, or lower environmental impact. Example: a more streamlined model rocket flies more accurately.

Gr 7 · Unit 7 · #28 · 3.5.6-8.JJ

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GR 7 · UNIT 7 STEELS 3.5.6-8.U #29

Principles and elements of design

The basic parts (elements, like shape and color) and guidelines (principles, like balance) used to create and judge the quality of a design.

Gr 7 · Unit 7 · #29 · 3.5.6-8.U

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GR 7 · UNIT 7 STEELS 3.5.6-8.U #30

How can you tell if one design is better than another?

Compare the strengths and weaknesses of each design using the principles and elements of design, then explain your reasons.

Gr 7 · Unit 7 · #30 · 3.5.6-8.U

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GR 7 · UNIT 7 STEELS 3.5.6-8.O #31

Accuracy

How close information or a measurement is to the true value. A water-purity tester gives fairly easy-to-check data.

Gr 7 · Unit 7 · #31 · 3.5.6-8.O

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GR 7 · UNIT 7 STEELS 3.5.6-8.O #32

Credibility

How trustworthy and believable a source of information is.

Gr 7 · Unit 7 · #32 · 3.5.6-8.O

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GR 7 · UNIT 7 STEELS 3.5.6-8.O #33

Relevance

How closely information connects to the question or problem you are working on.

Gr 7 · Unit 7 · #33 · 3.5.6-8.O

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GR 7 · UNIT 7 STEELS 3.5.6-8.O #34

Why is public opinion data hard to judge for accuracy?

Opinions can differ greatly from group to group, unlike readings from a physical measuring device.

Gr 7 · Unit 7 · #34 · 3.5.6-8.O

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GR 7 · UNIT 7 STEELS 3.5.6-8.X #35

Why should engineers defend their design decisions?

Explaining and backing up choices with evidence builds flexible thinking and accountability, and helps others understand the design.

Gr 7 · Unit 7 · #35 · 3.5.6-8.X

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GR 7 · UNIT 7 STEELS 3.5.6-8.X #36

Metacognition

Being aware of and understanding your own thought processes, like noticing how you solved a problem.

Gr 7 · Unit 7 · #36 · 3.5.6-8.X

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GR 7 · UNIT 7 STEELS 3.5.6-8.X #37

Habits of mind (in engineering)

Ways of thinking engineers practice until they become habits, like predicting outcomes early, thinking flexibly, and reflecting on their own thinking.

Gr 7 · Unit 7 · #37 · 3.5.6-8.X

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GR 7 · UNIT 7 STEELS 3.5.6-8.DD #38

Research and development (R&D)

A process of systematic testing and refining used to create new inventions or improve existing ones.

Gr 7 · Unit 7 · #38 · 3.5.6-8.DD

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GR 7 · UNIT 7 STEELS 3.5.6-8.DD #39

How has the light bulb changed since 1879?

Edison's 1879 bulb burned for only about 13 hours. Many tests, design changes, and innovations since then have made bulbs last much longer.

Gr 7 · Unit 7 · #39 · 3.5.6-8.DD

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Grade 7 · Unit 8 · Technology, Systems, and Society — Flashcards

PA STEELS 3.5.6-8.A, AA, B, BB, C, CC, D, E, EE, F, FF, G, GG, H, HH, I, J, K, KK, L, LL, Y, Z

GR 7 · UNIT 8 | STEELS 3.5.6-8.Y | #1

Science

The systematic investigation of the natural world.

Gr 7 · Unit 8 · #1 · 3.5.6-8.Y

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GR 7 · UNIT 8 | STEELS 3.5.6-8.Y | #2

Technology

Any change people make to the environment to meet their needs and wants. Examples: a bridge, a phone, a water filter.

Gr 7 · Unit 8 · #2 · 3.5.6-8.Y

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GR 7 · UNIT 8 | STEELS 3.5.6-8.Y | #3

Engineering

The process of creating or changing technologies. It is limited by physical laws, cultural norms, and economic resources.

Gr 7 · Unit 8 · #3 · 3.5.6-8.Y

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GR 7 · UNIT 8 STEELS 3.5.6-8.Y #4

How do science, technology, engineering, and math work together?

Each field adds something to building technological systems, and they overlap. Example: a design project may use science facts, math measurements, and engineering steps.

Gr 7 · Unit 8 · #4 · 3.5.6-8.Y

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GR 7 · UNIT 8 STEELS 3.5.6-8.BB #5

How do other school subjects help in designing technology?

Art skills help draw and show designs. Science helps too: Newton's laws of motion are used to build and test roller coasters or rockets.

Gr 7 · Unit 8 · #5 · 3.5.6-8.BB

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GR 7 · UNIT 8 STEELS 3.5.6-8.BB #6

Why study the history of technology?

It lets people learn from past successes and challenges when designing new products and systems.

Gr 7 · Unit 8 · #6 · 3.5.6-8.BB

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GR 7 · UNIT 8 STEELS 3.5.6-8.KK #7

Invention

A brand-new product, process, or idea that did not exist before. Example: Kevlar, a super-strong fiber.

Gr 7 · Unit 8 · #7 · 3.5.6-8.KK

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GR 7 · UNIT 8 STEELS 3.5.6-8.KK #8

Innovation

An improvement to an existing product, process, or system, or a new use for it. Example: LED bulbs improving on older bulbs.

Gr 7 · Unit 8 · #8 · 3.5.6-8.KK

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GR 7 · UNIT 8 STEELS 3.5.6-8.KK #9

Invention vs. innovation

An invention is something new that did not exist before. An innovation improves or finds a new use for something that already exists.

Gr 7 · Unit 8 · #9 · 3.5.6-8.KK

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GR 7 · UNIT 8 STEELS 3.5.6-8.KK #10

Creativity (in technology)

Using knowledge and experience from different subjects to make something new or to use something in a new way.

Gr 7 · Unit 8 · #10 · 3.5.6-8.KK

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GR 7 · UNIT 8 STEELS 3.5.6-8.KK #11

Can inventions happen by accident?

Yes. Stephanie Kwolek was looking for a replacement for steel cords in tires when she accidentally invented Kevlar.

Gr 7 · Unit 8 · #11 · 3.5.6-8.KK

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GR 7 · UNIT 8 STEELS 3.5.6-8.G #12

How does its context shape an invention or innovation?

Economic, political, cultural, and environmental conditions affect how a technology is designed and whether people accept it.

Gr 7 · Unit 8 · #12 · 3.5.6-8.G

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GR 7 · UNIT 8 STEELS 3.5.6-8.G #13

Why did LED bulbs largely replace older light bulbs?

People wanted lighting that was more efficient, lasted longer, and was better for the environment than incandescent or compact fluorescent bulbs.

Gr 7 · Unit 8 · #13 · 3.5.6-8.G

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GR 7 · UNIT 8 STEELS 3.5.6-8.CC #14

How did natural resources affect early human shelters?

Geography and the natural resources available in a place often decided what materials people used to build shelters.

Gr 7 · Unit 8 · #14 · 3.5.6-8.CC

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GR 7 · UNIT 8 STEELS 3.5.6-8.CC #15

What historical factors affect how technology develops?

Things like geography, which natural resources were available, and people's needs at that time in history.

Gr 7 · Unit 8 · #15 · 3.5.6-8.CC

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GR 7 · UNIT 8 STEELS 3.5.6-8.F #16

How can technology change the way people live?

Some technologies change how people think, interact, live, and communicate. Examples: smartphones and social media.

Gr 7 · Unit 8 · #16 · 3.5.6-8.F

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GR 7 · UNIT 8 STEELS 3.5.6-8.F #17

Why might a technology fit one society but not another?

Different societies have different needs, resources, and ways of life, so a technology can be appropriate and sustainable in one place but not another.

Gr 7 · Unit 8 · #17 · 3.5.6-8.F

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GR 7 · UNIT 8 STEELS 3.5.6-8.Z #18

How do technological systems connect with other systems?

They interact with economic, environmental, and social systems. Example: a delivery truck's GPS sends data to the company and alerts customers.

Gr 7 · Unit 8 · #18 · 3.5.6-8.Z

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GR 7 · UNIT 8 STEELS 3.5.6-8.Z #19

What drives improvements in technology?

Economic, political, social, and cultural parts of society push people to improve products, processes, and systems.

Gr 7 · Unit 8 · #19 · 3.5.6-8.Z

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GR 7 · UNIT 8 STEELS 3.5.6-8.FF #20

System

A group of parts that work together to reach a goal. Examples: a transportation system or a communication system.

Gr 7 · Unit 8 · #20 · 3.5.6-8.FF

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GR 7 · UNIT 8 STEELS 3.5.6-8.FF #21

Systems thinking

Looking at how every part of a system relates to the others, and how the whole system interacts with its environment and other systems.

Gr 7 · Unit 8 · #21 · 3.5.6-8.FF

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GR 7 · UNIT 8 STEELS 3.5.6-8.FF #22

What are two ways to analyze a system?

Look at its individual parts (like the parts of one computer), or look at the whole system and how it relates to others (like a computer network).

Gr 7 · Unit 8 · #22 · 3.5.6-8.FF

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GR 7 · UNIT 8 STEELS 3.5.6-8.EE #23

What do all technological systems have in common?

They are designed to reach goals, they turn inputs into outputs through processes, and they all use energy in some form.

Gr 7 · Unit 8 · #23 · 3.5.6-8.EE

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GR 7 · UNIT 8 STEELS 3.5.6-8.EE #24

Input

The resources that flow into a technological system. Examples: fuel going into a car, or a person flipping a light switch.

Gr 7 · Unit 8 · #24 · 3.5.6-8.EE

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GR 7 · UNIT 8 STEELS 3.5.6-8.EE #25

Process

The actions a system takes to change inputs into an output. Example: electricity flowing through wires to a bulb.

Gr 7 · Unit 8 · #25 · 3.5.6-8.EE

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GR 7 · UNIT 8 STEELS 3.5.6-8.EE #26

Output

The result of a technological system. It can have positive or negative impacts. Example: light filling a room.

Gr 7 · Unit 8 · #26 · 3.5.6-8.EE

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GR 7 · UNIT 8 STEELS 3.5.6-8.EE #27

Feedback

Information used to monitor or control a system. Example: a car's fuel gauge tells the driver when to add fuel.

Gr 7 · Unit 8 · #27 · 3.5.6-8.EE

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GR 7 · UNIT 8 STEELS 3.5.6-8.GG #28

Open-loop system

A system with no feedback path that needs a person to control it. Example: a light switch stays on until someone flips it off.

Gr 7 · Unit 8 · #28 · 3.5.6-8.GG

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GR 7 · UNIT 8 STEELS 3.5.6-8.GG #29

Why is a room's light switch an open-loop system?

It has no feedback. A person flips the switch (input), electricity flows to the bulb (process), and the room lights up (output).

Gr 7 · Unit 8 · #29 · 3.5.6-8.GG

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GR 7 · UNIT 8 STEELS 3.5.6-8.HH #30

Closed-loop system

A system with a feedback path that controls itself automatically, with no person needed. Example: a home water heater.

Gr 7 · Unit 8 · #30 · 3.5.6-8.HH

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GR 7 · UNIT 8 STEELS 3.5.6-8.HH #31

How does a thermostat make a water heater closed-loop?

The thermostat senses the water temperature (feedback) and automatically turns the heater on or off as needed.

Gr 7 · Unit 8 · #31 · 3.5.6-8.HH

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GR 7 · UNIT 8 STEELS 3.5.6-8.HH #32

Open-loop vs. closed-loop system

An open-loop system has no feedback and needs a person to control it. A closed-loop system uses feedback to control itself.

Gr 7 · Unit 8 · #32 · 3.5.6-8.HH

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GR 7 · UNIT 8 STEELS 3.5.6-8.K #33

Sensor

A device that detects something, like light, heat, or motion, and sends that information to control a system. Robots use sensors.

Gr 7 · Unit 8 · #33 · 3.5.6-8.K

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GR 7 · UNIT 8 STEELS 3.5.6-8.K #34

Automation

Using machines or computer controls to do tasks without a person doing each step. Example: a robot on an assembly line.

Gr 7 · Unit 8 · #34 · 3.5.6-8.K

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GR 7 · UNIT 8 STEELS 3.5.6-8.K #35

Algorithmic thinking

Planning a clear sequence of steps to solve a problem. It is used to create and test automated solutions.

Gr 7 · Unit 8 · #35 · 3.5.6-8.K

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GR 7 · UNIT 8 STEELS 3.5.6-8.K #36

Safety device

A part on a machine that protects the user from getting hurt. Example: a guard that covers a moving blade.

Gr 7 · Unit 8 · #36 · 3.5.6-8.K

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GR 7 · UNIT 8 STEELS 3.5.6-8.LL #37

Do all technologies use the same processes?

No. Different technologies use different sets of processes. Data processing and construction each have their own steps.

Gr 7 · Unit 8 · #37 · 3.5.6-8.LL

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GR 7 · UNIT 8 STEELS 3.5.6-8.LL #38

Data processing

Working with information: designing, summarizing, storing, retrieving, reproducing, evaluating, and communicating it.

Gr 7 · Unit 8 · #38 · 3.5.6-8.LL

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GR 7 · UNIT 8 STEELS 3.5.6-8.LL #39

What processes are part of construction?

Designing, developing, evaluating, making and producing, marketing, and managing.

Gr 7 · Unit 8 · #39 · 3.5.6-8.LL

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GR 7 · UNIT 8 STEELS 3.5.6-8.AA #40

Technology transfer

Adapting an existing product, system, or process to solve a problem in a different setting.

Gr 7 · Unit 8 · #40 · 3.5.6-8.AA

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GR 7 · UNIT 8 STEELS 3.5.6-8.AA #41

What is an example of technology transfer?

An automatic pump built for the Mars Viking space probe was changed into an insulin pump that helps patients control blood sugar.

Gr 7 · Unit 8 · #41 · 3.5.6-8.AA

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GR 7 · UNIT 8 STEELS 3.5.6-8.AA #42

Microcontroller

A tiny computer on a single chip that can be programmed to control a device. Students can use one to solve design problems.

Gr 7 · Unit 8 · #42 · 3.5.6-8.AA

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GR 7 · UNIT 8 STEELS 3.5.6-8.A #43

Maintenance

Taking care of a product or system so it keeps working properly. Example: cleaning a bike chain.

Gr 7 · Unit 8 · #43 · 3.5.6-8.A

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GR 7 · UNIT 8 STEELS 3.5.6-8.A #44

Troubleshooting

Finding and fixing the cause of a problem in a product or system, step by step. Many manuals give troubleshooting tips.

Gr 7 · Unit 8 · #44 · 3.5.6-8.A

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GR 7 · UNIT 8 STEELS 3.5.6-8.A #45

Where can you learn how to use and fix a product?

Written and graphical information, like manuals and diagrams, explains how to use a product and check if it works properly.

Gr 7 · Unit 8 · #45 · 3.5.6-8.A

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GR 7 · UNIT 8 STEELS 3.5.6-8.J #46

Diagnose (a system)

To figure out what is causing a problem in a system before adjusting or repairing it.

Gr 7 · Unit 8 · #46 · 3.5.6-8.J

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GR 7 · UNIT 8 STEELS 3.5.6-8.J #47

Why learn safety procedures before using tools?

Tools and machines can be dangerous. Safety procedures, learned through teaching and demonstration, help you diagnose, adjust, and repair safely.

Gr 7 · Unit 8 · #47 · 3.5.6-8.J

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GR 7 · UNIT 8 STEELS 3.5.6-8.J #48

CNC machine

A computer numerically controlled machine: a tool, like a lathe, that a computer controls to cut materials. Worn parts need adjusting.

Gr 7 · Unit 8 · #48 · 3.5.6-8.J

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GR 7 · UNIT 8 STEELS 3.5.6-8.B #49

How can instruments help judge a product?

Instruments measure how well an everyday product performs, giving evidence to decide if it meets its criteria and constraints.

Gr 7 · Unit 8 · #49 · 3.5.6-8.B

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GR 7 · UNIT 8 STEELS 3.5.6-8.B #50

Photovoltaic system

A system of solar panels that changes sunlight directly into electricity.

Gr 7 · Unit 8 · #50 · 3.5.6-8.B

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GR 7 · UNIT 8 STEELS 3.5.6-8.B #51

Rated output

The amount a product is designed to produce. Measuring a solar panel system's power shows if it is working at its rated output.

Gr 7 · Unit 8 · #51 · 3.5.6-8.B

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GR 7 · UNIT 8 STEELS 3.5.6-8.L #52

Performance test

A test that measures how well a product works, such as its accuracy or speed.

Gr 7 · Unit 8 · #52 · 3.5.6-8.L

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GR 7 · UNIT 8 STEELS 3.5.6-8.L #53

Destructive testing

Testing a material until it bends or breaks to measure its strength and durability.

Gr 7 · Unit 8 · #53 · 3.5.6-8.L

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GR 7 · UNIT 8 STEELS 3.5.6-8.L #54

How can you gather data about a technological system?

Design a test or use a device, like a water or air quality tester, and use digital tools to collect and analyze the data.

Gr 7 · Unit 8 · #54 · 3.5.6-8.L

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GR 7 · UNIT 8 STEELS 3.5.6-8.C #55

Simulation

Using a digital model of a system to see what happens. Changing the model shows how outcomes would change.

Gr 7 · Unit 8 · #55 · 3.5.6-8.C

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GR 7 · UNIT 8 STEELS 3.5.6-8.C #56

How could choosing a different design change the outcomes?

Focusing on different criteria could lead to different effects on people, cultures, and the environment, positive or negative.

Gr 7 · Unit 8 · #56 · 3.5.6-8.C

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GR 7 · UNIT 8 STEELS 3.5.6-8.I #57

Can a technology be helpful and harmful at the same time?

Yes. Flush toilets improved health, but they also created a need for water treatment that uses lots of energy and fresh water.

Gr 7 · Unit 8 · #57 · 3.5.6-8.I

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GR 7 · UNIT 8 STEELS 3.5.6-8.I #58

What shapes the form and function of a technology?

The criteria designers choose when developing it. Form is a product's shape; function is what it does.

Gr 7 · Unit 8 · #58 · 3.5.6-8.I

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GR 7 · UNIT 8 STEELS 3.5.6-8.E #59

What effects can a new technology have on society?

Positive and negative effects, both expected and unexpected. Example: a new phone may help people stay in touch but create electronic waste.

Gr 7 · Unit 8 · #59 · 3.5.6-8.E

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GR 7 · UNIT 8 STEELS 3.5.6-8.H #60

Why do technology decisions need careful compromise?

Every technology has benefits and consequences, and people see them from different perspectives, so competing factors must be balanced.

Gr 7 · Unit 8 · #60 · 3.5.6-8.H

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GR 7 · UNIT 8 STEELS 3.5.6-8.H #61

Design for obsolescence

Designing products to wear out or go out of date so people buy new ones. It adds to unsustainable rates of consumption.

Gr 7 · Unit 8 · #61 · 3.5.6-8.H

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GR 7 · UNIT 8 STEELS 3.5.6-8.D #62

Renewable resource

A resource that can be replaced naturally in a fairly short time if used carefully. Examples: trees, crops.

Gr 7 · Unit 8 · #62 · 3.5.6-8.D

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GR 7 · UNIT 8 STEELS 3.5.6-8.D #63

Non-renewable resource

A resource that cannot be replaced in a human lifetime once it is used up. Examples: oil, coal, metal ores.

Gr 7 · Unit 8 · #63 · 3.5.6-8.D

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GR 7 · UNIT 8 STEELS 3.5.6-8.D #64

Inexhaustible resource

A resource that will not run out no matter how much people use it. Examples: sunlight and wind.

Gr 7 · Unit 8 · #64 · 3.5.6-8.D

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GR 7 · UNIT 8 STEELS 3.5.6-8.D #65

How do people get resources from Earth?

By methods such as harvesting (crops, trees), mining (rocks, metals), and drilling (oil, natural gas).

Gr 7 · Unit 8 · #65 · 3.5.6-8.D

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GR 7 · UNIT 8 STEELS 3.5.6-8.D #66

By-product

Something extra made while producing or getting something else, often waste. Example: rock left over from mining.

Gr 7 · Unit 8 · #66 · 3.5.6-8.D

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GR 7 · UNIT 8 STEELS 3.5.6-8.D #67

Sustainable resource use

Using resources in a way that meets people's needs now without using them up or harming the environment for the future.

Gr 7 · Unit 8 · #67 · 3.5.6-8.D

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GR 7 · UNIT 8 STEELS 3.5.6-8.D #68

How can technology cause environmental challenges?

Making and using technology uses up resources and creates waste. Some choices trade off the environment against the economy.

Gr 7 · Unit 8 · #68 · 3.5.6-8.D

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GR 7 · UNIT 8 STEELS 3.5.6-8.E #69

Reduce

Using less of something so less waste is made. Example: printing on both sides of paper.

Gr 7 · Unit 8 · #69 · 3.5.6-8.E

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GR 7 · UNIT 8 STEELS 3.5.6-8.E #70

Reuse

Using an item again, for the same or a new purpose, instead of throwing it away. Example: refilling a water bottle.

Gr 7 · Unit 8 · #70 · 3.5.6-8.E

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GR 7 · UNIT 8 STEELS 3.5.6-8.E #71

Recycle

Breaking down used materials so they can be made into new products. Example: turning old cans into new aluminum.

Gr 7 · Unit 8 · #71 · 3.5.6-8.E

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